

CONTROLLING MEMBRANE PROTEIN BIOGENESIS AND FUNCTION THROUGH REVERSIBLE LIPID MODIFICATION

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ABSTRACT

Protein S-acylation is a widespread and reversible post-translational modification that regulates proteins across virtually all cellular processes, yet its remarkable functional versatility remains underappreciated. Here, I will illustrate how cycles of S-acylation and deacylation can accompany a membrane protein throughout its lifetime, controlling its folding, trafficking and ultimately its function. As a model, I will discuss CMG2/ANTXR2, a collagen VI receptor required for extracellular matrix homeostasis that is also hijacked as the major cellular receptor for anthrax toxin. We find that sequential S-acylation events promote CMG2 folding by protecting folding intermediates from ER quality control, its subsequent trafficking through the secretory pathway, and its delivery to the plasma membrane. There, ligand-induced deacylation enables cytoskeletal remodeling and receptor endocytosis. Thus, the same reversible lipid modification performs distinct functions as CMG2 progresses through the cell. Finally, exploiting this regulatory cycle pharmacologically strongly attenuates anthrax toxin toxicity in vivo. Together, these findings illustrate how dynamic S-acylation can provide spatial and temporal control over multiple stages in the life of a protein, from its birth to its function in health and disease.